

ARANYABAHUGUNA

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Education

Vellore Institute of Technology
B. Tech CSE(Core)

CGPA: 8.77

Expected June, 2026

Technical Skills

- **Languages:** Java, JavaScript, SQL
- **Frontend:** React.js, Tailwind CSS, Bootstrap, HTML, CSS
- **Backend:** Node.js, Express.js, REST APIs, JWT, RBAC
- **Databases:** MongoDB, PostgreSQL (Supabase), MySQL, Firebase
- **Tools & Platforms:** Git, GitHub, Postman, AWS (Basic), Figma.

Projects

SOS-Service

Node.js | Express.js | React.js | Supabase (PostgreSQL)

Sep. 24 – Feb. 25

- Architected a full-stack solution leveraging React.js and Node.js/Express to enable real-time emergency event handling and seamless front-end & backend synchronization.
- Engineered data-driven infrastructure using Supabase (PostgreSQL) and REST APIs, incorporating Google Maps and Geo-Hashing for precise location-based tracking and visualization.
- Integrated YOLOv8 based machine learning services into the backend pipeline to automate emergency detection and fire severity classification from incoming visual data.
- Strengthened system-wide security by implementing AES-256 encryption and Role-Based Access Control (RBAC) to ensure end-to-end protection for sensitive data.

Suno Samjho

Node.js | Express.js | React.js | Supabase (PostgreSQL)

Sep. 25 – Mar. 25

- Full Stack Development (MERN): Developed a scalable mental health web application using MongoDB, Express.js, React, and Node.js, implementing secure authentication, REST APIs, and dynamic user dashboards for real-time interaction.
- Frontend & User Experience: Built responsive and intuitive React-based UI with real-time data visualization, ensuring smooth user interaction, accessibility, and engaging mental health tracking features.
- Backend & Database Architecture: Designed and deployed backend services with Node.js and Express, integrating Neon DB (serverless PostgreSQL) for efficient data management, ensuring high performance, scalability, and secure data handling.
- AI & Multimodal System: Built a multilingual AI-driven mental health screening system leveraging NLP models (IndicBERT) and speech processing (Whisper ASR) to analyze voice and text inputs for depression, anxiety, and stress detection.

Inventory Management

MongoDB | Node.js | Express.js | React.js

Jan. 25 – Feb. 25

- Architected and developed a comprehensive MERN-stack infrastructure, building a responsive React.js frontend and a high performance Node.js/Express backend to manage end-to-end inventory lifecycles.
- Engineered a scalable data layer and communication bridge by designing optimized MongoDB schemas and RESTful APIs, ensuring seamless data flow and real-time synchronization between the UI and database.
- Hardened system security by implementing robust authentication and authorization protocols using JWT, crypt, and custom RBAC middleware.
- Enhanced database performance through advanced schema modelling and query optimization, significantly reducing latency and ensuring data consistency across complex inventory transactions.

Hackathon and Internship

- **HPCL Hackathon (Semi-Finalist):** Developed a web-based portal for Hindustan Petroleum Corporation Limited to monitor port operations, enabling real-time tracking of logistics, vessel status, and operational workflows.
- **AI for Bharat Hackathon (Semi-Finalist):** Built an AI-powered text-to-video generation system for social media services, automating content creation and enhancing user engagement through dynamic visual outputs.
- **Indian Space Lab – Drone Technology Intern:** Gained hands-on experience with UAV design, flight control systems, and autonomous navigation, with exposure to mission planning tools and aerial surveying applications.

Extra-Curricular and Achievements

- Runner-up, Ecom-Thon: Pitched a tech-based solution at a competitive tech-cultural fest.
- Core Member, NSS VIT Bhopal: Completed 240+ hours of community service; awarded the prestigious NSS 'B' Certificate.
- Founder & Coordinator, Pahadi Club VIT Bhopal.
- Sports: Cricket – Winner (Aarambh Intercollege 2022), Runner-up (Advitiya Intercollege 2023).

Additional Information

- Hobbies: Playing Guitar, Gaming, Cooking, Writing, Sports.
- Languages: English (Professional working proficiency), Hindi (Native or bilingual proficiency).